

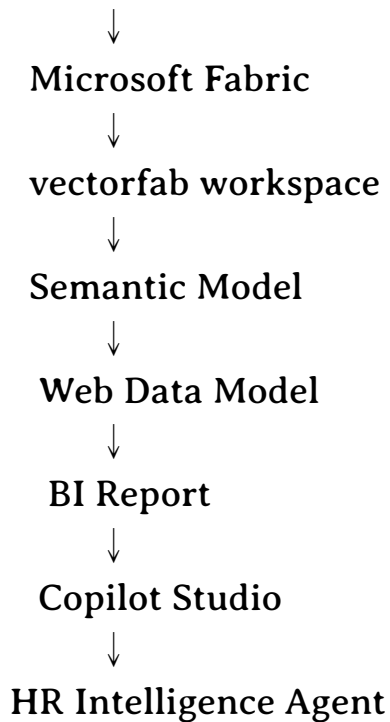
Lab 1 - Enterprise HR Intelligence Agent using Fabric

Platform

- Microsoft Fabric portal
 - Existing **vectorfab** Power BI Pro workspace
 - Fabric semantic model
 - BI web report
 - Copilot Studio
 - [Contoso HR Employee Data.csv](#)
-

Architecture

Contoso_HR_Employee_Data.csv



PART 1 — PARTICIPANT EXECUTION

Step 1 — Open Fabric

Open:

[Microsoft Fabric](#)

Sign in with the assigned lab account.

Validation

You should see:

Workspaces



vectorfab

Open: vectorfab

Do not create a new workspace.

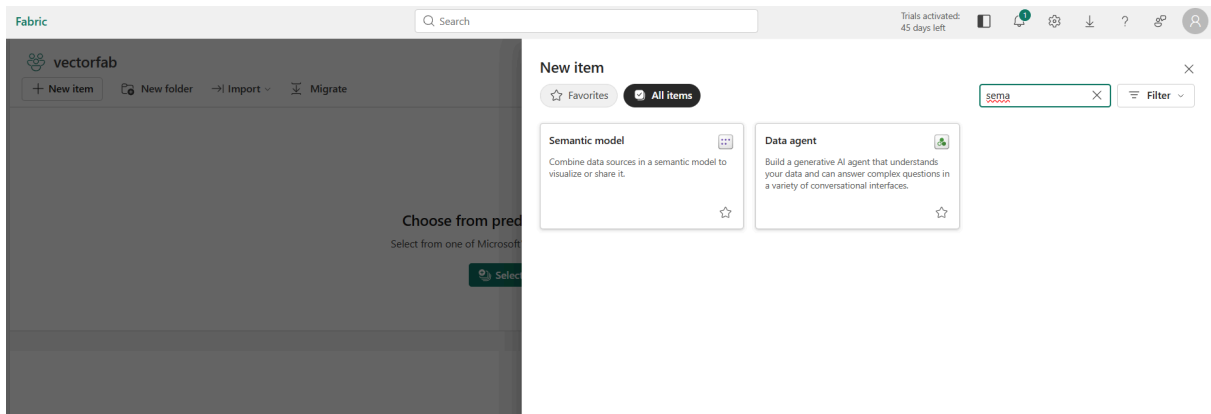
Step 2 — Create the Semantic Model Directly From CSV

Inside: vectorfab

Select: + New item

Search:

Semantic model

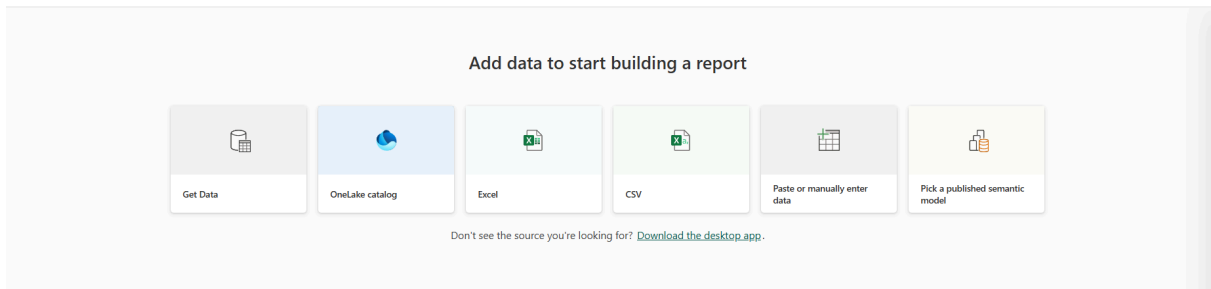


Step 3 — Select CSV

In the window:

Select a data source

Choose: **CSV**



You should now get the CSV import screen.

Select: **Upload file**

Choose:

Contoso_HR_Employee_Data.csv

Power Query

Get data

Connect to data source

Text/CSV

File

[Learn more](#)

Connection settings

Link to file

Upload file

Contoso_HR_Employee_Data.csv

Upload successful (4.3 KB)

Connection credentials

Connection

https://labskraft27-my.sharepoint.com/personal/christi...

Authentication kind: Organizational account

[Edit connection](#)

Cancel

Next

Step 4 — Preview Data

Fabric will display the CSV preview.

Power Query

Get data

Preview file data

URL: https://labskraft27-my.sharepoint.com/personal/christinefabric_labskraft_com/Documents/Apps/Microsoft Power Query/Uploaded Files/Contoso_HR_Employee_Data.csv

File origin

65001: Unicode (UTF-8)

Delimiter

Comma

Data type detection

Based on first 200 rows

EmployeeID	Name	Department	Salary	JoiningDate
101	Anita Sharma	Finance	65000	1/15/2024
102	Rahul Mehta	IT	72000	2/10/2024
103	Priya Singh	HR	58000	3/5/2024
104	Amit Patel	IT	75000	3/20/2024
105	Sneha Rao	Marketing	60000	4/12/2024
106	Arjun Verma	Finance	67000	5/8/2024
107	Kavita Iyer	Sales	55000	6/18/2024
108	Vikram Das	Operations	63000	7/22/2024
109	Pooja Nair	Sales	59000	8/14/2024
110	Saurabh Gupta	IT	77000	9/3/2024
111	Suresh Sharma	IT	93500	8/29/2020
112	Divya Kumar	Operations	71500	5/3/2021
113	Poonam Yadav	Operations	72000	6/11/2020
114	Nitin Joshi	Marketing	73000	11/4/2022
115	Alok Bansal	Finance	64500	11/4/2022
116	Preeti Garg	Sales	62000	9/29/2023
117	Tarun Reddy	HR	62500	4/13/2021
118	Neha Mukherjee	IT	86500	2/4/2021
119	Ritu Chatterjee	Operations	69000	8/26/2023

Back

Cancel

Next

Verify these columns:

EmployeeID

Name

Department

Salary

JoiningDate

Verify that the records look correct.

Select: **Next**

Microsoft's current workflow includes a preview/validation stage before creating the semantic model. ([Microsoft Learn](#))

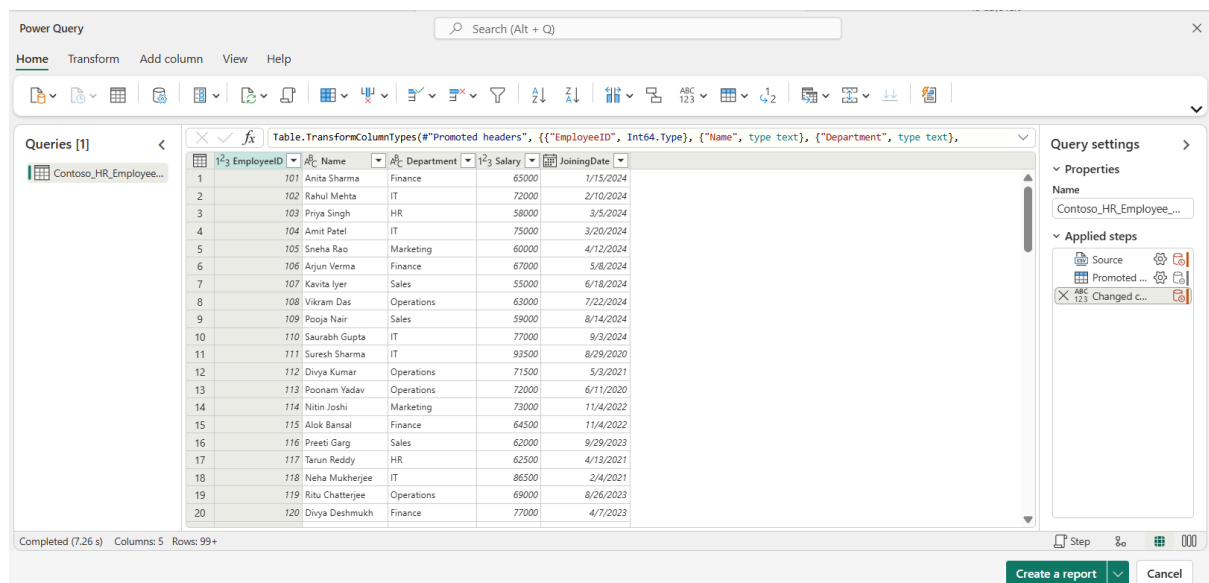
Step 5 — Create the Semantic Model

You should now see options similar to:

Create a report / Create a semantic model only

For this lab choose:

Create a semantic model only



This gives us the semantic model and the web report in one flow.

Name the semantic model:

Contoso_HR_Intelligence

Select: **Create**



This folder is empty

Create, save, or move your items to this folder

Semantic model name

Contoso_HR_Intelligence

Create

Cancel

Note : It May Takes upto 5 minutes to deploy so please wait (close the tab and reopen to validate in workspace)

Step 6 — Verify the Semantic Model

Return to: **vectorfab**

You should now have something similar to:

Contoso_HR_Intelligence

The CSV data is imported into the Power BI semantic model. Microsoft confirms that CSV imports create model tables in the Power BI service.

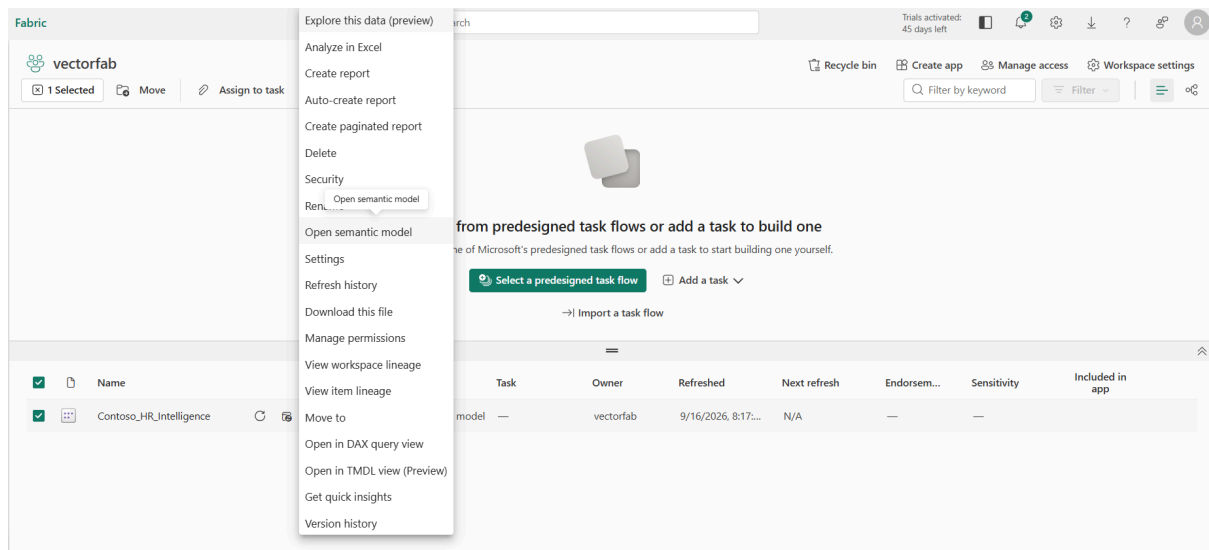
Step 7 — Open the Data Model

From:

Contoso_HR_Intelligence

select:

... → Open data model



This opens the browser-based modeling experience.

Microsoft documents browser-based editing of semantic models in the Fabric workspace. ([Microsoft Learn](#))

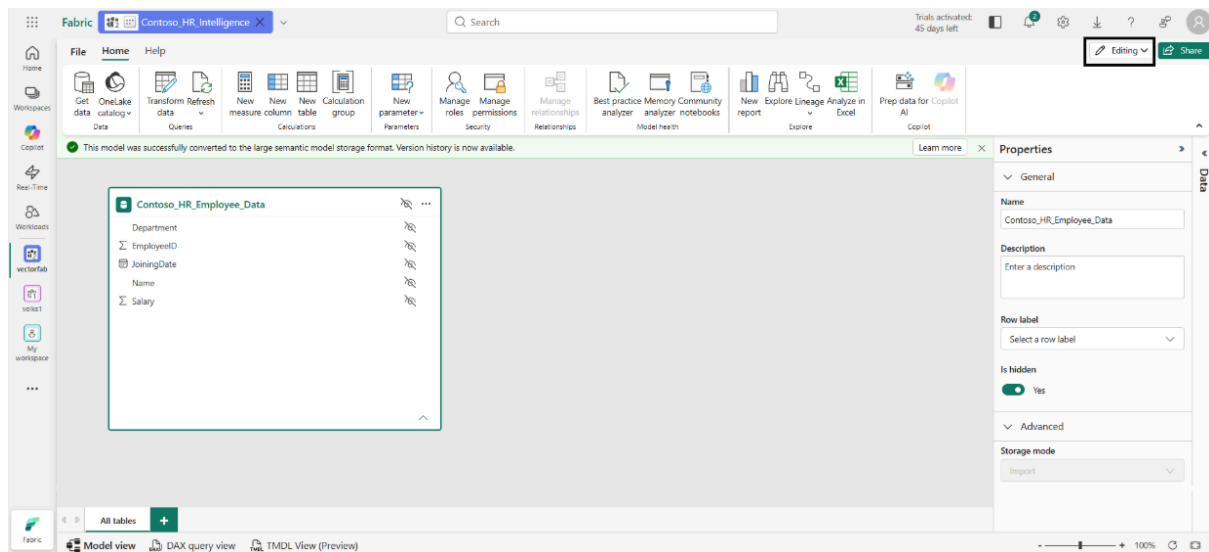
Step 8 — Verify Employee Table

In the model view, verify:

Employee

- EmployeeID
- Name
- Department
- Salary

JoiningDate



Make Sure we are in editing mode for validating data types.

Step 9 — Configure EmployeeID

Select:

Employee → EmployeeID

Set the data type to:

Whole number

This is our employee identifier.

Step 10 — Configure Salary

Select:

Employee → Salary

Set:

Data type → Whole number

Set the appropriate:

Currency format

Step 11 — Configure JoiningDate

Select:

Employee → JoiningDate

Set:

Data type → Date

Step 12 — Create Total Salary Measure

Select the **Employee** table.

Choose: **New measure**

Enter:

Total Salary = SUM(Employee[Salary])

Save it.

Step 13 — Create Employee Count

Create another measure:

Employee Count = COUNTROWS(Employee)

Step 14 — Create Average Salary

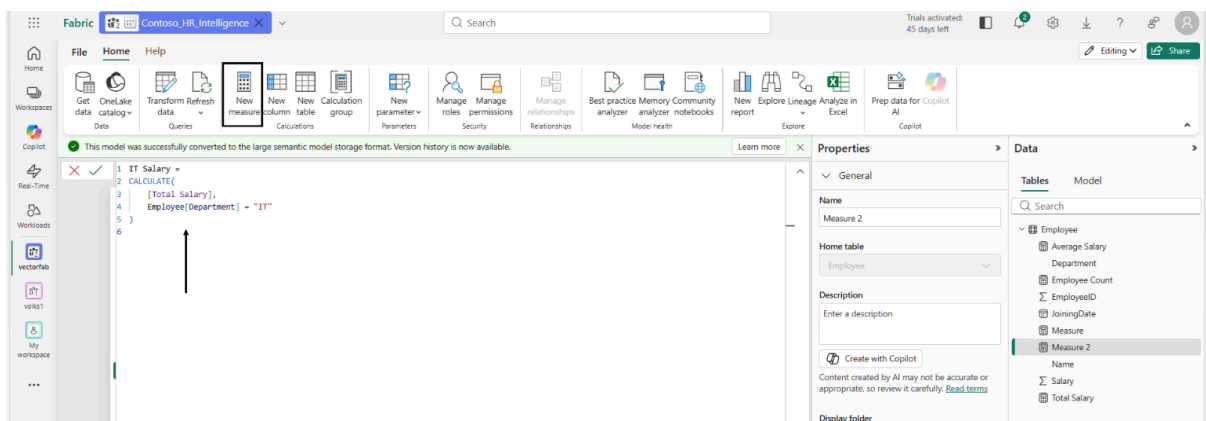
Create:

Average Salary = AVERAGE(Employee[Salary])

Step 15 — Create IT Salary

Create:

IT Salary =
CALCULATE(
[Total Salary],
Employee[Department] = "IT"
)



Step 16 — Validate Measures

Before creating the report, validate the measures.

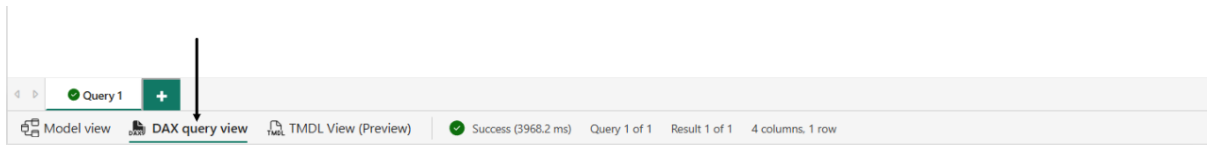
Step 1 — Use DAX Query View

At the bottom of your screen, click:

DAX query view

You currently have:

Model view | DAX query view | TMDL View



Step 2 — Run a Validation Query

Paste this:

EVALUATE

ROW(

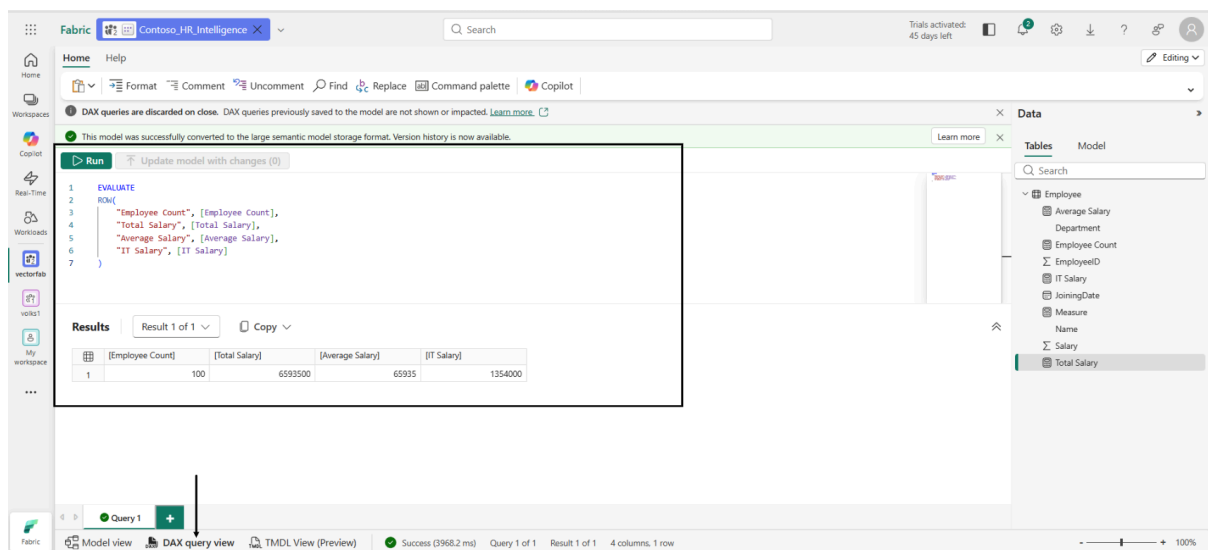
"Employee Count", [Employee Count],

"Total Salary", [Total Salary],

"Average Salary", [Average Salary],

"IT Salary", [IT Salary]

)



Then click **Run**.

Expected:

Measure	Expected
Employee Count	100
Total Salary	6593500
Average Salary	65,395
IT Salary	135400

These values are our lab ground truth.

Step 3 — Validate Department Breakdown

After the first query succeeds, run:

EVALUATE

SUMMARIZECOLUMNS(

Employee[Department],

"Employee Count", [Employee Count],

"Total Salary", [Total Salary],

"Average Salary", [Average Salary]

)

ORDER BY

Employee[Department]

Home Help

Format Comment Uncomment Find Replace Command palette Copilot

DAX queries are discarded on close. DAX queries previously saved to the model are not shown or impacted. [Learn more](#)

This model was successfully converted to the large semantic model storage format. Version history is now available. [Learn more](#)

Run Update model with changes (0)

```
1 EVALUATE
2 SUMMARIZECOLUMNS(
3     Employee[Department],
4     "Employee Count", [Employee Count],
5     "Total Salary", [Total Salary],
6     "Average Salary", [Average Salary]
7 )
8 ORDER BY
9     Employee[Department]
```

Results Result 1 of 1 Copy

	Employee[Departme...	[Employee Count]	[Total Salary]	[Average Salary]
1	Finance	13	901000	69307.69
2	HR	21	1264000	60190.48
3	IT	17	1354000	79647.06
4	Marketing	14	915500	65392.86
5	Operations	16	1042000	65125
6	Sales	19	1117000	58789.47

Query 1 +

Step 4 — Check the Source Data

You can also run:

EVALUATE
Employee

and click **Run**.

You should see the 100 employee records.

This confirms:

CSV → Semantic Model → Measures → Query

PART 2 — CREATE THE HR REPORT IN FABRIC

Step 17 — Open Report

From:

Contoso_HR_Intelligence

select:

... → **Open report**

or:

New report

depending on the current workspace UI.

The report can be created directly in the Power BI web experience; Microsoft documents creating reports from semantic models in the workspace.

Step 18 — Create Salary KPI

Add a:

Card

Use: Total Salary

Expected: 651,000

Step 19 — Create Employee KPI

Add another:

Card

Use: Employee Count

Expected: 100

Step 20 — Create Average Salary KPI

Add:

Card

Use: Average Salary

Expected: 65,100

Step 21 — Create IT Salary KPI

Add:

Card

Use: IT Salary

Expected: 224,000

Step 22 — Department Analysis

Add a:

Column chart

X-axis

Department

Y-axis

Employee Count

You should see:

Finance

HR
IT
Marketing
Operations
Sales

Step 23 — Employee Detail Table

Add a:

Table

Fields:

EmployeeID
Name
Department
Salary
JoiningDate

This provides the detailed HR view.

Step 24 — Save the Report

Save the report as:

Contoso_HR_Intelligence_Report

PART 3 — SEMANTIC MODEL BUSINESS CONTEXT

Step 25 — Add Model Description

Open:

Semantic model → Settings

Add the following description where your current UI provides the semantic-model description field:

Contoso HR employee analytics model.

Employee contains employee identity, department, salary and joining-date information.

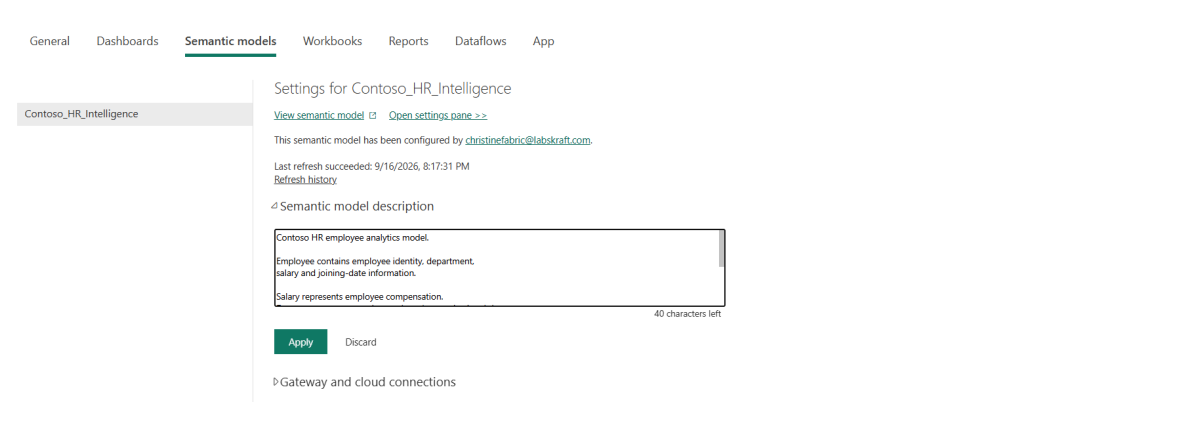
Salary represents employee compensation.

Department represents the employee's organizational department.

Use the predefined measures for salary calculations.

Do not infer missing employee information.

Power BI semantic models are specifically intended to provide business-friendly terminology, metrics and representation for analysis.



PART 4 — REALTIME VALIDATION IN FABRIC

Before we touch Copilot Studio, the Fabric-only portion must pass.

Step 26 — Validate Total Salary

Ask/use the report to confirm:

Total Salary = 6593500

Step 27 — Validate IT

Filter:

Department = IT

Expected:

Employees = 17

Salary = 1264000

Step 28 — Validate Finance

Filter:

Department = Finance

Expected:

Employees = 13

Salary = 901000

Step 29 — Validate Sales

Filter:

Department = Sales

Expected:

Employees = 19

Salary = 1117000

PART 5 — COPILOT STUDIO

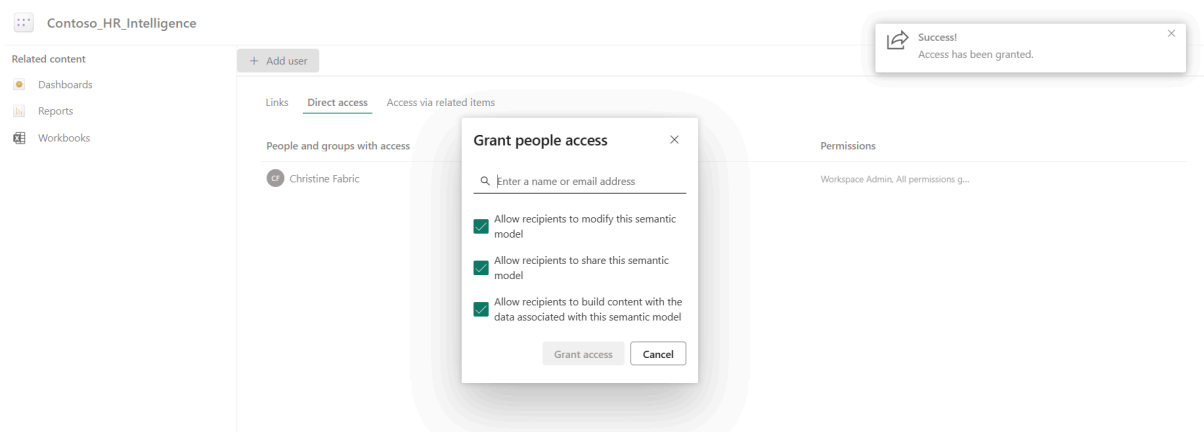
Now we leave Fabric only because the agent is created in Copilot Studio.

Step 30 — Open Copilot Studio

Open:

[Microsoft Copilot Studio](#)

Use the same organizational account.



Step 31 — Create Agent

Select:

Create → New agent / Blank agent

Name:

Contoso HR Intelligence Agent

Description:

An enterprise HR analytics assistant that answers questions using the approved Contoso HR semantic model.

Step 32 — Configure Agent Instructions

Use:

You are Contoso's HR Intelligence Agent.

Answer employee analytics questions using the connected Contoso HR data source.

Use the predefined business measures whenever possible.

Do not invent employee records, salaries, departments, or dates.

If information is unavailable in the connected data, clearly state that it is unavailable.

For numerical questions, use the connected data source rather than estimating.

Workflow — HR Natural Language → DAX → Power BI → Answer

Architecture

User Question



When an agent calls the flow



Agent 1 — NLP → DAX



Power BI — Run a query against a dataset



Agent 2 — Result → Natural Language



Respond to the agent

Step 1 — Open Copilot Studio

Go to:

Create a new agent.

Name

Contoso HR Intelligence Agent

Description

An enterprise HR analytics assistant that answers employee and salary questions using the Contoso HR semantic model.

For the agent instructions, use:

You are the Contoso HR Intelligence Agent.

Answer HR analytics questions using the connected Contoso HR data source.

Use the Employee table and the following measures when appropriate:

- Employee Count
- Total Salary
- Average Salary
- IT Salary

Known business fields:

- EmployeeID
- Name
- Department
- Salary
- JoiningDate

Always provide factual answers based on the connected data.

Do not invent employee information or departments that are not present in the data.

If the requested information is unavailable, clearly state that it is not available.

PART 6 — Create the Agent Workflow

Step 1 — Open Copilot Studio

Go to:

Copilot Studio → Contoso HR Intelligence Agent

Open:

Build → Tools

Select:

Add a tool → Workflow

A new workflow designer opens. Microsoft supports creating an agent flow with the required trigger and response action.

PART 7 — Configure the Trigger

Step 2 — Trigger

Use:

When an agent calls the flow

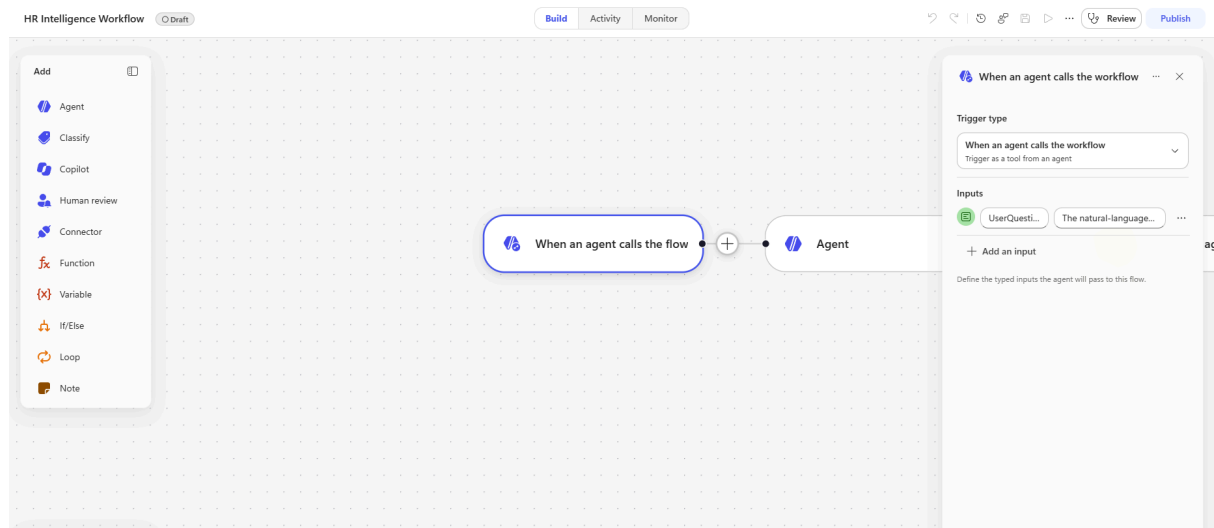
Step 3 — Add input

Open the trigger.

Add an input:

Property	Value
Type	Text
Name	UserQuestion
Description	The natural-language HR analytics question asked by the user.

The trigger now receives the user's question.



PART 8 — Agent 1: NLP → DAX

Step 4 — Add Agent

Click:

+ → Add an action → Agent

Microsoft's current Agent node supports dynamic content from earlier workflow steps and exposes its response to subsequent actions.

Step 5 — Agent instructions

Paste:

You are an HR analytics query-generation agent.

Your job is to interpret the user's natural-language HR analytics question and convert it into a valid DAX query for the Contoso_HR_Intelligence semantic model.

Available table:

Employee

Available columns:

- EmployeeID
- Name
- Department
- Salary
- JoiningDate

Available measures:

- Employee Count
- Total Salary
- Average Salary
- IT Salary

Rules:

1. Only use the Employee table and the columns and measures listed above.
2. Never invent tables, columns, measures, employees, or departments.
3. Return only a valid DAX query beginning with EVALUATE.
4. For employee-count questions, use the [Employee Count] measure.
5. For total-salary questions, use the [Total Salary] measure.
6. For average-salary questions, use the [Average Salary] measure.
7. For IT salary questions, use the [IT Salary] measure.
8. For department-specific questions, apply a filter on Employee[Department].
9. For employee-specific questions, filter using Employee[Name].
10. If the question cannot be answered using the available schema, return exactly:

UNSUPPORTED

Examples:

Question:

How many employees are in IT?

Return:

```
EVALUATE
ROW(
    "Employee Count",
    CALCULATE(
        [Employee Count],
        Employee[Department] = "IT"
    )
)
```

Question:

What is the total salary for Finance?

Return:

```
EVALUATE
ROW(
    "Total Salary",
    CALCULATE(
        [Total Salary],
        Employee[Department] = "Finance"
    )
)
```

Question:

What is Rahul Mehta's salary?

Return:

```

EVALUATE
ROW(
    "Salary",
    CALCULATE(
        MAX(Employee[Salary]),
        Employee[Name] = "Rahul Mehta"
    )
)

```

The user's question is:

Step 6 — Insert dynamic input

Immediately after:

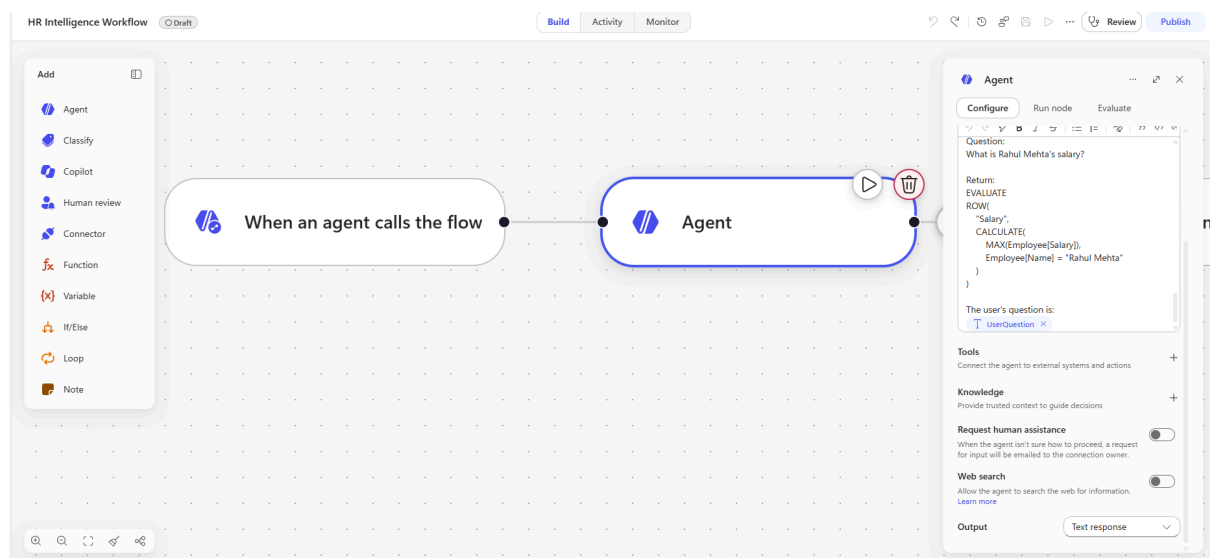
The user's question is:

insert the blue **UserQuestion** dynamic token.

It should visually appear as:

The user's question is:

[UserQuestion]



Do not type [UserQuestion] manually.

Step 7 — Agent output

Set:

Output → Text response

Do not select Structured output.

The result must be the DAX text. The Agent node's response can then be consumed by later workflow actions.

PART 9 — Query Power BI

Step 8 — Add Power BI action

Click:

+

Select:

Power BI → Run a query against a dataset

Step 9 — Configure workspace

Set:

Workspace:

vectorfab

Step 10 — Configure dataset

Set:

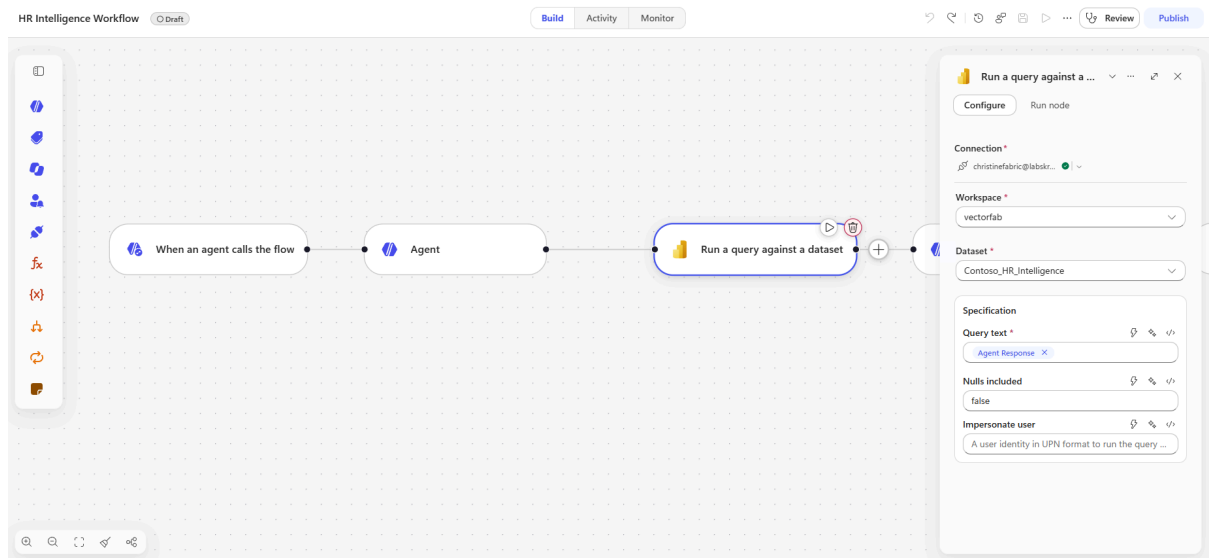
Dataset:

Contoso_HR_Intelligence

Step 11 — Configure Query text

For Query text, insert:

Agent → Agent Response



Step 12 — Nulls

Set:

Nulls included = false

Leave:

Impersonate user = blank

unless your organization's security design specifically requires impersonation.

PART 10 — Agent 2: Result → Answer

Step 13 — Add second Agent

After Power BI:

+ → Add an action → Agent

Name it:

HR Answer Formatter

Step 14 — Agent 2 instructions

Use:

You are an HR analytics answer formatter.

Use ONLY the result returned by the Power BI query.

Answer the user's original question in concise natural language.

Do not invent values.

Do not perform unsupported calculations.

If no rows are returned, state that no matching data was found.

Original user question:

Insert:

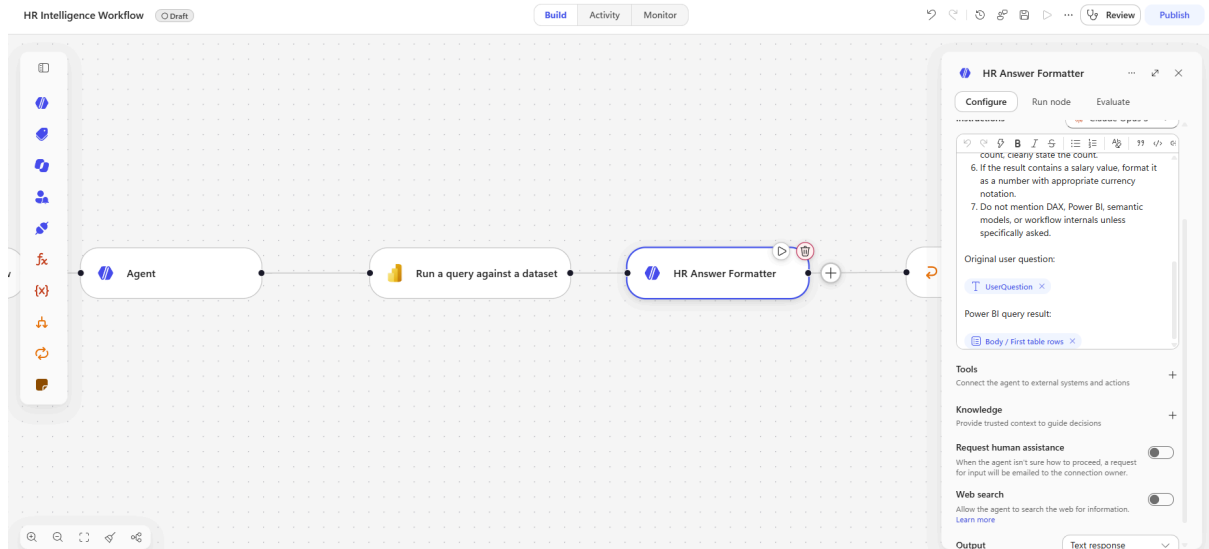
UserQuestion

Then add:

Power BI query result:

Insert:

First table rows



So the final structure is:

Original user question:
[UserQuestion]

Power BI query result:
[First table rows]

Step 15 — Agent 2 settings

Set:

Output = Text response

Keep:

Web search = Off

Request human assistance = Off

PART 11 — Respond to Agent

Step 16 — Add response

At the end of the workflow:

+ → **Respond to the agent**

Microsoft requires this response action for a workflow that is going to be used as an agent tool.

Step 17 — Add output

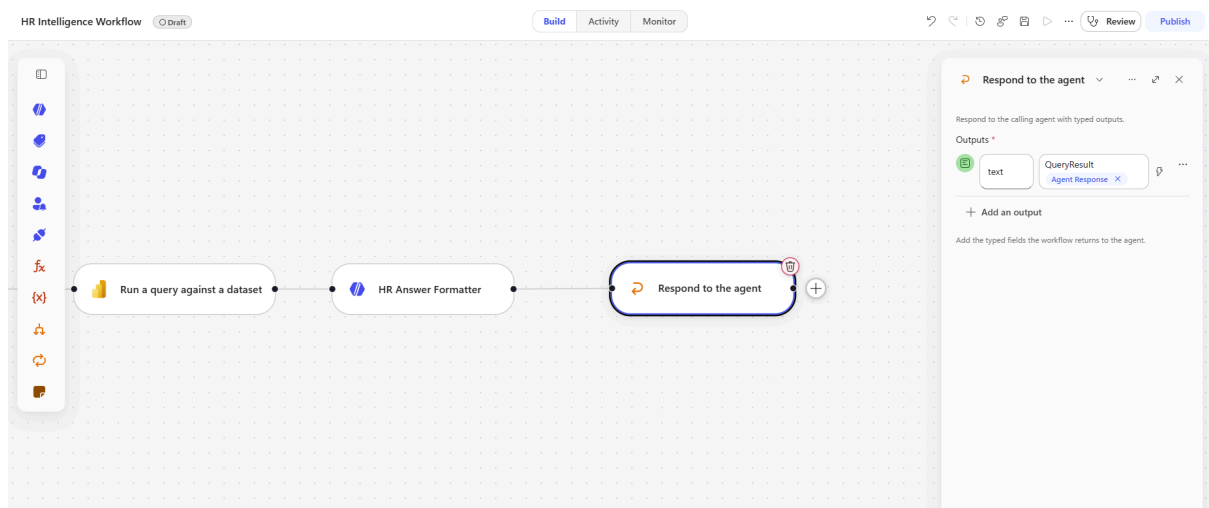
Add:

Text

Name: Answer

For the value, select:

Agent 2 → Agent Response



PART 12 — Publish

Step 18 — Publish the workflow

Select:

Publish

The workflow must be published before it becomes available for adding to the agent as a workflow tool.

PART 13 — Validate the Workflow

Step 19 — Test Agent 1

Use:

How many employees are in IT?

Expected Agent 1 behavior:

Natural language



DAX

For example:

EVALUATE

ROW(

 "Employee Count",

 CALCULATE(

 [Employee Count],

 Employee[Department] = "IT"

)

)

Step 20— Test Power BI

Verify that the Power BI action succeeds.

For your current 100-row dataset, the returned count should reflect your current data.

Step 21 — Test Agent 2

Verify that it converts the returned Power BI row into a normal sentence.

Example format:

There are 17 employees in the IT department.

The exact number depends on your current 100-row dataset.

PART 14 — Add Workflow to HR Agent

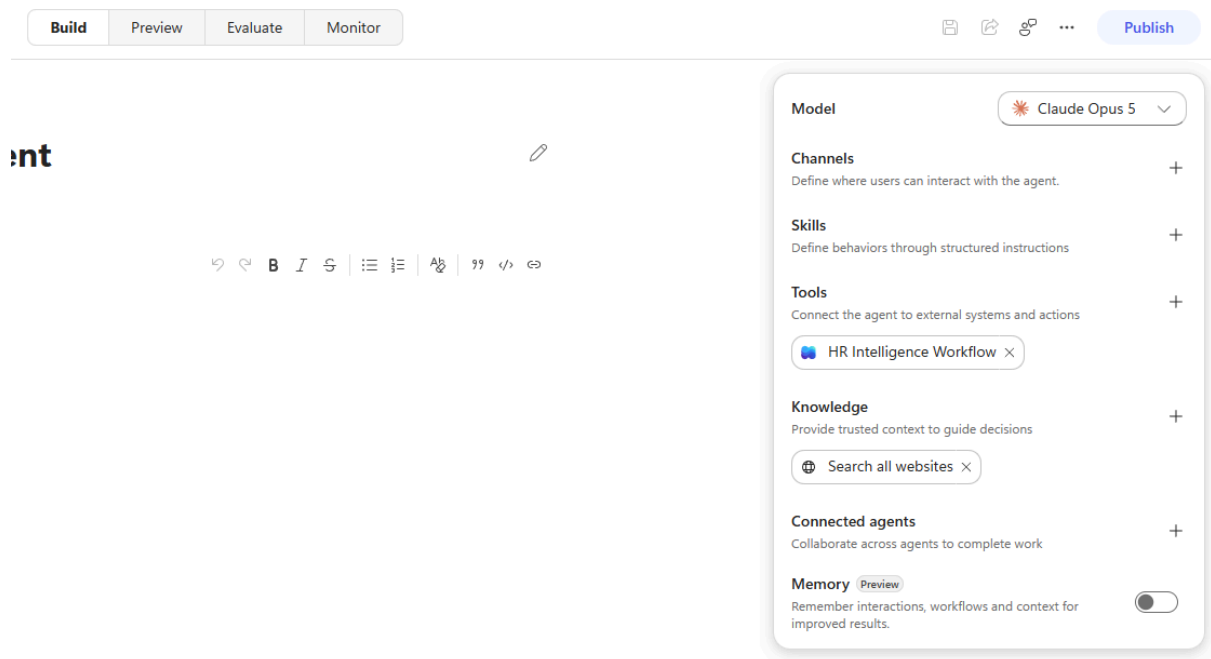
Step 22 — Return to Agent

Go back to:

Contoso HR Intelligence Agent → Build → Tools

Select:

Add a tool → Workflows



Find your published workflow.

Select: Add

Microsoft lists published workflows that have the required trigger and response action in this area.

PART 15 — FINAL AGENT VALIDATION

Once the connection is configured:

Test 1

Ask: How many employees are there?

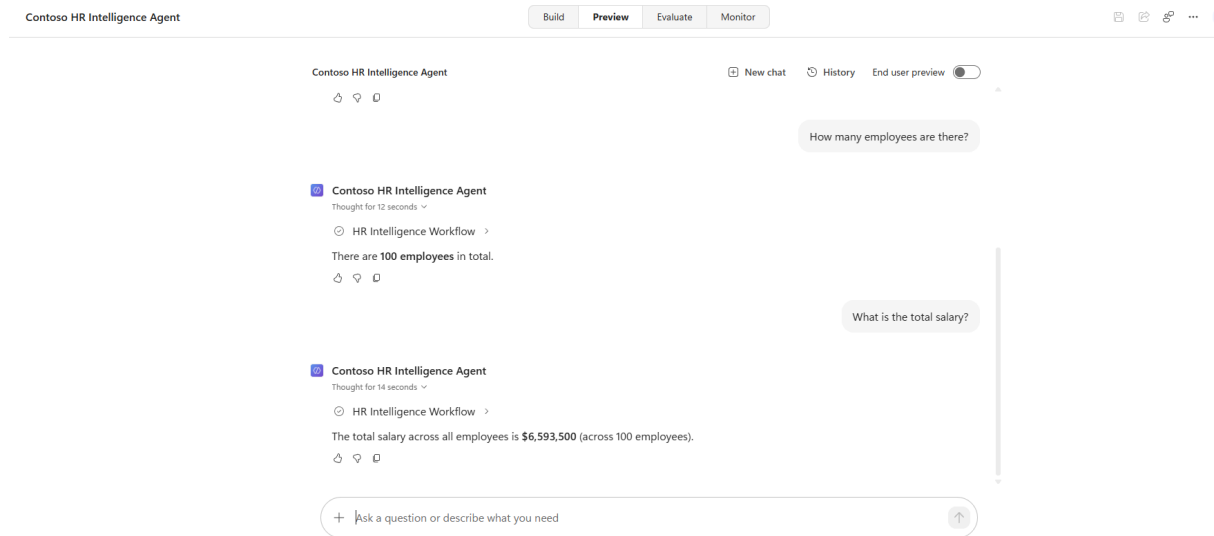
Expected: 100

Test 2

Ask:

What is the total salary?

Expected: 6,593,500



Test 3

Ask:

What is the average salary?

Expected: 65,100

Test 4

Ask:

How many employees work in IT?

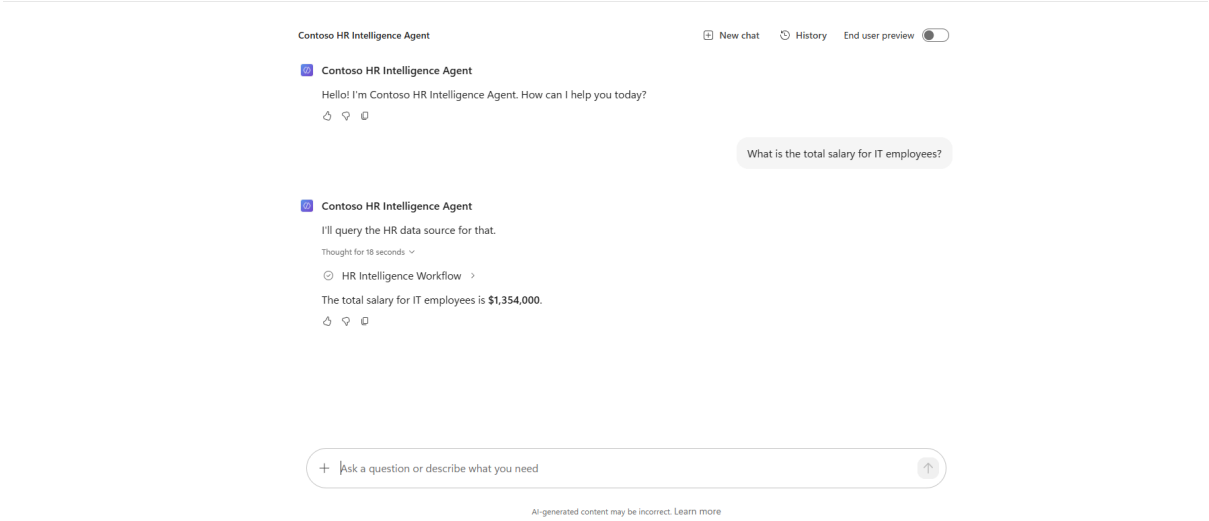
Expected: 17

Test 5

Ask:

What is the total salary for IT employees?

Expected: \$1,354,000



Test 6

Ask:

Where does Rahul Mehta work?

Expected:

IT

Test 7 — Grounding Test

Ask:

How many employees work in the Legal department?

